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ABSTRACT

Methane emissions during the rice-growing season primarily originate from the rhizosphere. Understanding the impact of different soil pore structures on methane emissions can deepen our insight into the pore-scale mechanisms of greenhouse gas (GHG) emissions in paddy soils and assist in developing management practices to mitigate these emissions. In this study, we investigated the differences in pore characteristics between rhizosphere and non-rhizosphere soils using x-ray micro-computed tomography. Subsequently, we employed the three-dimensional (3D) lattice Boltzmann method (LBM) to simulate tortuosity (using single-phase LBM) and methane flow and retention (using multi-phase LBM) in both rhizosphere and non-rhizosphere soils. Our findings indicate that the pore structure and hydraulic properties of rhizosphere soil were different from those of non-rhizosphere soil due to the role of plant roots. Due to increased connectivity and decreased tortuosity, the methane saturation in rhizosphere soil (0.2–0.25) is lower than that of non-rhizosphere soil ($\sim$0.35), reflecting a reduced methane retention capacity of rhizosphere soil. From the perspective of pore size distribution, we observed a reduction in the proportion of small pores (80–200 μm) and increase in the critical diameter in rhizosphere soil, contributing to its lower methane retention capacity. Our study highlights the structural differences and methane retention capacities between rhizosphere and non-rhizosphere soils, establishing a preliminary relationship between them. This provides a new perspective for studying GHG release at the pore scale.

INTRODUCTION

Tackling climate change related to anthropogenic greenhouse gas (GHG) emissions is a global challenge. Methane (CH_4), the second largest greenhouse gas next to carbon dioxide, has attracted considerable attention.[1] Globally, 50%–65% of methane emissions originate from human activities, surpassing contributions from natural sources like wetlands.[2] Anthropogenic methane emissions come from fossil fuel use, agriculture, waste management, and industrial activities.[3] Notably, 22% of agricultural methane emissions come from rice cultivation.[4] Methane emissions from rice fields can be reduced through proper fertilization, effective irrigation management, and carefully selecting rice varieties and fertilizers.[5]

Due to its high yield, which enhances weed control and promotes optimal plant growth, continuous flooding is commonly used in most rice-growing regions.[6] In contrast to rice–wheat rotations and traditional single-crop rice fields, continuously flooded fields maintain a persistent flood layer throughout most of the rice-growing season, including non-rice cultivation periods. Year-round flooding creates ideal anaerobic conditions for methane production. As a result, these fields emit substantial amounts of methane during the rice-growing season, with methane emission fluxes significantly higher than those in other rice field types.[7] Therefore, conducting comprehensive studies on methane emissions from continuously flooded rice fields is highly valuable.

Methane in flooded rice fields primarily escapes into the atmosphere through the aerenchyma of rice plants and by diffusing from soil water.[8] The movement of methane in the soil is predominantly influenced by pore structure characteristics.[9] For example, soil porosity and pore size play crucial roles in the emission process, significantly affecting the release of greenhouse gases from the soil.[10,11] Similarly, pore connectivity strongly influences gas diffusion: in poorly connected networks, gas diffusion is more constrained compared to well-connected networks, where diffusion resistance is primarily due to pore size distribution rather than network topology.[12] Gas migration in a texturally heterogeneous soil system (e.g., with a low-permeability clay lens embedded in a sandy formation) differs markedly from that in a homogeneous soil system due to texture- (or porosity-) induced tortuosity effects.[13] These observations highlight how soil pore characteristics impact gas emissions from the soil.

Rice is a plant with fibrous roots. Previous research indicates that porosity decreases by 22%–24% near fibrous root systems.[14,15] Root systems create macropores through root channel formation and the processes of expansion and contraction.[16] Consequently, roots alter the physical properties of the surrounding soil, which in turn influences methane emissions.[17]

Currently, most studies on methane emissions from paddy fields focus on soil types, farming methods, and microbial activities. Methane emission is a dynamic process, influenced by the spatial and temporal variabilities of soil parameters.[18,19] This creates significant challenges in studying methane emissions from soils. In contrast to these widely conducted macro-scale studies, very few have evaluated methane emissions at the micro-scale.[20] The role of soil properties at the pore scale in methane emissions urgently requires exploration, and corresponding research methods at this scale need to be developed. However, the complexity of solid boundaries and the highly heterogeneous nature of rhizosphere soil pore structures present significant challenges for computational fluid dynamics methods when describing the flow within the pores.[21–23] Consequently, the lattice Boltzmann method (LBM) has gained popularity for its ability to handle complex boundaries.[24,25] LBM is particularly adept at processing large numbers of fluid cells in computed tomography (CT) images and managing complex solid boundary conditions, with graphics processing unit (GPU) acceleration—a parallel computing architecture optimized for high-throughput tasks—significantly enhancing simulation speed.[26] This approach has enabled precise simulations of soil permeability and its anisotropic characteristics[27] and has also provided accurate predictions of the air–water two-phase flow, facilitating the study of water–air distribution in a three-dimensional (3D) pore space.[28,29] So far, gas migration and the behavior of methane in pores have been explored through LBM simulations,[30,31] and study on the relationship between greenhouse gas and pore structures extracted from CT imaging has also been conducted.[32] The pore structures of the soil can be accurately characterized using x-ray micro-computed tomography (mCT), which provides a foundation for LBM-based simulations of methane emissions at the pore scale.

As mentioned above, current research on methane emission from soil primarily focuses on large-scale factors, such as soil types, farming practices, and microbial activities, with limited exploration of the mechanisms by which the soil pore structure influences methane emissions. However, the intricate and porous structure of soil poses considerable challenges for computational fluid dynamics methods. Therefore, there is an urgent need for a method that can effectively simulate methane emissions from soil, enabling a detailed investigation of the impact of soil structure on methane emission at the pore scale.

We hypothesize that differences in pore structure—such as pore size distribution, porosity, connectivity, and tortuosity—between rhizosphere and non-rhizosphere soils will lead to variations in methane emissions. To test this hypothesis, our study integrates x-ray mCT and GPU-accelerated LBM simulations, presenting a promising approach to investigate the pore-scale mechanisms of methane emissions in rhizosphere soil, with a focus on how pore architecture influences these emissions.

MATERIALS AND METHODS

X-ray micro-computed tomography data acquisition and image processing

Soil samples, with a volumetric water content of approximately 65%, were collected from a permanently flooded rice field located in Zhejiang Province, southern China (28°11′23″N, 121°1′10″E). Specifically, root-free soil samples and root-containing soil samples were collected. To maintain the soil structure's integrity, we used a ring cutter to collect undisturbed samples, which were then securely enclosed in sealed ring pipes. Each sample was wrapped in plastic to prevent moisture loss and transported to the laboratory in an aluminum box for protection. Prior to CT scanning, the undisturbed samples were equilibrated in a pressure membrane apparatus set at 33 kPa. Each sample had a diameter and height of 5 cm. Soil images were captured using x-ray computed tomography (CT) with a resolution of 40 μm. Root-free soil samples were collected far from the rice roots and were treated as non-rhizosphere soil samples. Since most area in root-containing soil samples is located within 5 mm of the roots, it can be considered that the characteristics of these samples are predominantly influenced by the rhizosphere.[33–35] We treat these root-containing soil samples as rhizosphere soil samples. The non-rhizosphere soil samples were labeled RL1 and RL2, while the rhizosphere soil samples were labeled R1 and R2.

Image normalization, filtering, segmentation, cropping, and other preprocessing steps were performed using ImageJ software. Brightness normalization was performed with the enhanced contrast module in ImageJ to reduce brightness variations between different CT slices. For non-rhizosphere soil samples, we utilized the threshold segmentation method. Initially, the IsoData algorithm was employed to autonomously determine the threshold by clustering the internal gray values of the image.[36] We then manually reviewed each image to select the optimal threshold range. The soil x-ray images were segmented into two phases: pore and solid.

For rhizosphere soil samples, a two-step binary segmentation process was used to distinguish among soil pores, plant roots, and solid material. First, the original grayscale image was binarized, with white areas representing soil particles and black areas encompassing plant roots and soil pores. The resulting binary image served as a mask for a second binarization, which separated plant roots from soil pores. Due to significant noise in the images, repeated erosion, expansion, and noise filtering were necessary to achieve accurate results [Fig. 1(b)]. To determine the pore size distribution of each soil sample, we applied the maximum sphere method to extract pore radius parameters.

The 3D soil structure model in this study was reconstructed from segmented image slices. Using this method, we obtained images of both soil pores and extracted soil roots. Based on these images, we constructed complete 3D models, as shown in Fig. 2. Figures 2(a) and 2(b) present schematic diagrams of the pore structures of RL1 and RL2, respectively. Figures 2(c) and 2(d) display the reconstructed images of the roots and pores of R1 and R2, respectively. Figure 2 illustrates that the root and pore structure models were reconstructed following the image segmentation process. To minimize edge effects and optimize computational efficiency, the binary images were cropped to 550×550 pixels before further processing. These images were then stacked to generate the 3D structural model from the 2D images.

Pore structure analysis

In our study, we employed morphological models to assess pore size distribution.[37] We modeled the pore space as a combination of spheres and cylinders of varying diameters. For each pore, we defined its diameter as the diameter of the largest sphere that could fit within the pore. The identification of pores and the calculation of metrics, such as pore radius and volume, were performed using the "pnextract-master" algorithm.[38]

To quantify the connectivity of the pore network, the Euler number was utilized to characterize the connectivity of soil pores.[39,40] The Euler number (E) was determined by subtracting the number of redundant connections (C) from the count of disconnected bond clusters (I). The value of C represented the number of connections that could be removed without resulting in additional orphaned clusters. Consequently, the Euler number served as a metric for connectivity: a positive Euler number signifies poor connectivity, while a negative Euler number suggests better connectivity. Euler number values were obtained using the MORPHOLIBJ plugin for ImageJ.[41]

When characterizing the relationship between the pore structure and fluid flow, the critical diameter (d_c) provides a more accurate reflection of how pore architecture impacts fluid mobility.[42] Consequently, we employed the parallelized Particle Analyzer algorithm from the BoneJ plugin to identify individual percolating pore clusters within the binary images. Subsequently, we calculated the critical pore diameter using the SoilJ plugin.[41,43]

To assess the impact of soil pore geometry on water flow, we calculated the tortuosity of soil samples. Tortuosity is defined as the ratio between the actual distance traveled by water in the direction of the applied pressure difference and the length of the medium. Since tortuosity cannot be measured directly, it is typically calculated through image analysis or flow simulation. The soil and rice roots were treated as the solid phase, and the fluid simulations were conducted within the pore spaces. In this study, the tortuosity is determined using the fluid velocity field.[44] Specifically, it is calculated by dividing the average magnitude of the local flow velocity by the component of the velocity in the average flow direction, as expressed in Eq. (1). Here, $\langle \mathbf{u} \rangle$ represents the average magnitude of the local flow velocity, and $\langle u_j \rangle$ denotes the component of the velocity in the average flow direction. The resulting tortuosity value is indicated by Γ. The larger value of Γ indicates a more tortuous microchannel within the pores, whereas a value of 1 implies that the internal channel is a straight line

$$\Gamma = \frac{\langle \mathbf{u} \rangle}{\langle u_j \rangle}. \tag{1}$$

Fluid numerical modeling

To simulate methane emissions, we first converted the CT images into binary data files following postprocessing. These files were then used to define the geometry for the simulations, which were executed using "Multiphase Flow Lattice Boltzmann Method (MF-LBM)," a high-performance lattice Boltzmann method code.[45] Currently,

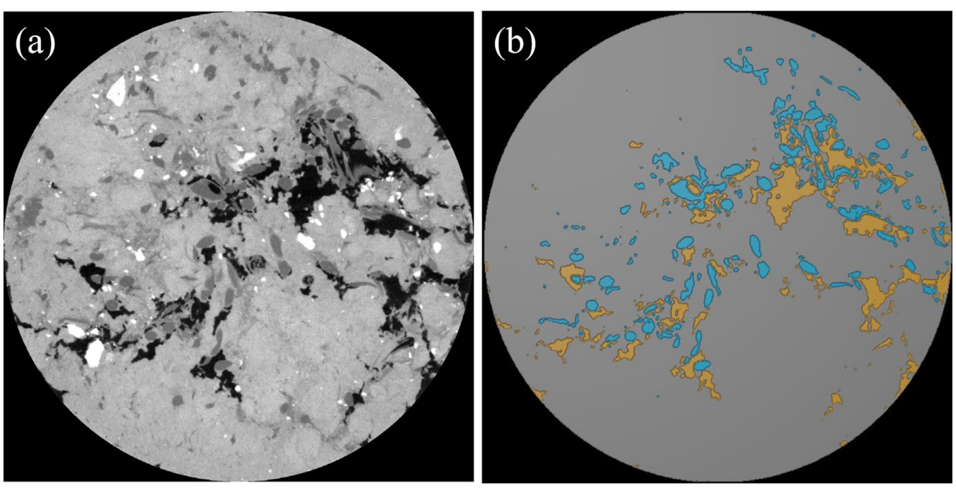

FIG. 1. Diagram of CT image processing: (a) original grayscale image of soil CT image with roots, (b) multi-color segmented image with four material classes: exterior areas in black, pores in yellow, roots in blue, and soil in gray.

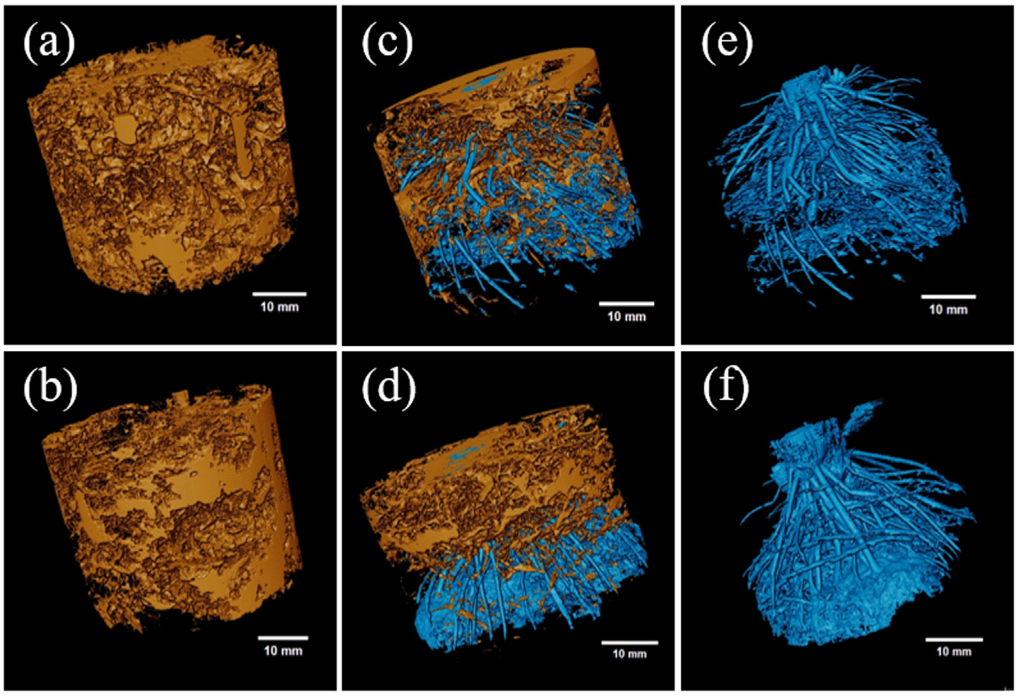

FIG. 2. Three-dimensional reconstruction of soil samples: (a) pore reconstruction of the non-rhizosphere soil sample (RL1), (b) pore reconstruction of the non-rhizosphere soil sample (RL2), (c) root and pore reconstruction of the rhizosphere soil sample (R1), and (d) root and pore reconstruction of the rhizosphere soil sample (R2). From (a) to (d), roots are depicted in blue and pores are shown in yellow. (e) and (f) illustrate the root reconstructions for the two rhizosphere soil samples.

three-dimensional lattice Boltzmann simulations commonly use discrete velocity models, such as D3Q15, D3Q19, and D3Q27, where D represents the calculated spatial dimension, and Q represents the number of spatial velocity directions. For this study, we selected the D3Q19 model due to its optimal balance between accuracy and computational efficiency. The D3Q19 model provides a suitable number of discrete velocity settings, ensuring both effective fluid flow calculations and efficient simulations. This model is well-suited for the porous structures required in our calculations, as it maintains high computational efficiency and accuracy for most fluid dynamic problems.[46] The macroscopic fluid density ρ and velocity u are defined as

$$\rho = \sum_{i=0}^{18} f_i, \tag{2}$$

$$u = \frac{1}{\rho}\sum_{i=0}^{18} f_i e_i, \tag{3}$$

where f is the distribution function and e_i represents the microscopic velocity along the ith direction. We employed a single-phase flow simulation to obtain the permeability of the pore structure.[47] The motion equation for the single-phase flow can be expressed as

$$f_i(x + e_i\delta t, t + \delta t) - f_i(x, t) = -\frac{1}{\tau}\left[f_i(x, t) - f_i^{eq}(\rho, u)\right] + \delta t \cdot S_i(x, t) \quad i = 0, 1, 2\ldots 18. \tag{4}$$

In the above equation, the left side of the equal sign is the propagation of fluid particles, and the right side of the equal sign is the collision process of fluid particles. Gravity is not considered, so $S_i(x, t)=0$, where the equilibrium distribution function is expressed as

$$f_i^{eq}(\rho, u) = w_i\rho(x)\left[1 + \frac{3e_i u}{c^2} + \frac{9(e_i u)^2}{2c^4} - \frac{3u^2}{2c^2}\right], \tag{5}$$

where f_i^{eq} is the equilibrium distribution function at position x and time t along the ith direction, and w_i is the weight coefficient of the D3Q19 model. The parameter c is defined by Eq. (6), where δx represents the lattice spacing and δt denotes the time step. For the D3Q19 model, $c_s^2 = \frac{c^2}{3}$, where c_s is the speed of sound. The dimensionless relaxation time τ, which is related to the kinematic viscosity ν is given by Eq. (7)

$$c = \frac{\delta x}{\delta t}, \tag{6}$$

$$\nu = \frac{1}{3}\left(\tau - \frac{1}{2}\right)\frac{\delta x^2}{\delta t}. \tag{7}$$

Permeability is a crucial soil hydraulic property closely related to the pore structure. The widely used Kozeny–Carman relationship offers an empirical equation that links permeability to porosity,[48] demonstrating a strong correlation between these two factors. To preliminarily assess the accuracy of the LBM in this study, we perform experimental measurements of the permeability of the microfluidic structure and compare the results with the simulated values. Single-phase flow simulations were performed using the LBM to calculate permeability of the microfluidic structure. The formula for calculating permeability is as follows:

$$k = \frac{\bar{\rho u} \nu L}{\Delta P}, \tag{8}$$

where $\bar{\rho u}$ is the average mass flux through the sample, ν is the kinematic viscosity of the fluid, L is the length of sample, and ΔP is the pressure difference between the two ends of the sample.

Then, we utilized a two-phase flow simulation method to model the gas emissions. The dual distribution function approach was used to represent the behaviors of different fluids, which were distinguished by their respective colors. Specifically, fluid r[49] and fluid b (blue) were represented by their corresponding particle distribution functions f_i^r and f_i^b. The mixture at position x and time t is then represented by $f_i(x,t) = f_i^r(x,t) + f_i^b(x,t)$. The lattice Boltzmann equations for both fluid r and fluid b, as described by Tölke,[50] are expressed as follows:

$$f_i^s(x + e_i\delta t, t+\delta t) = f_i^s(x,t) + \Omega_i^{s(3)}\left\{\Omega_i^{s(1)} + \Omega_i^{s(2)}\right\}, s = r, b, \quad i = 0, \ldots, 18, \tag{9}$$

where the superscript s indicates fluid r or fluid b, and $\Omega_i^{s(1)}$, $\Omega_i^{s(2)}$, and $\Omega_i^{s(3)}$ are the collision operators responsible for the viscous effect, surface tension effect, and separation of different fluids, respectively. $\Omega_i^{s(1)} = -\mathbf{M}^{-1}\mathbf{S}[(\mathbf{Mf}) - \mathbf{m}^{eq}]$ and $\Omega_i^{s(2)} = \delta_t \mathbf{M}^{-1}\hat{\mathbf{F}}$, where $\hat{\mathbf{F}}$ is obtained from $\Omega_i^{s(2)} = \delta_t \mathbf{M}^{-1}\left(\mathbf{I} - \frac{1}{2}\mathbf{S}\right)\mathbf{M}\bar{\mathbf{F}}$. The evolution equation based on the multi-relaxation time LBM framework is established

$$\begin{aligned} &f_i^s(\mathbf{x}+\mathbf{e}_i\delta t, t+\delta t) \\ &\quad = f_i^s(\mathbf{x},t) + \Omega_i^{s(3)}\left\{-\mathbf{M}^{-1}\mathbf{S}[(\mathbf{Mf}) - \mathbf{m}^{eq}] + \delta_t\mathbf{M}^{-1}\hat{\mathbf{F}}\right\}, \\ &\quad \mathrm{s} = \mathrm{r}, \mathrm{b}, \quad i = 0, \ldots, 18, \end{aligned} \tag{10}$$

where $\mathbf{M}$ represents the transformation matrix,[51] $\mathbf{m}^{eq}$ is the vector of equilibrium moments, and $\mathbf{S}$ is the diagonal collision matrix, as defined in Eq. (11). To separate the two fluids from the fluids in the mixture, we applied the recoloring operator to the post-collision volume using the R–K recoloring scheme described by Latva-Kokko and Rothman.[52] In this study, the recoloring parameter in the R–K recoloring scheme was set to 0.9 to achieve a relatively clear interface

$$\mathbf{S} = diag(0, s_e, s_\xi, 0, s_q, 0, s_q, 0, s_q, s_\nu, s_\pi, s_\nu, s_\pi, s_\nu, s_\nu, s_\nu, s_m, s_m, s_m), \tag{11}$$

where s_ν is related to the fluid viscosity. Building on the work of Chen *et al.*,[53] we identified a set of relaxation rates that performed well in our study: $s_e = 1.19$, $s_\xi = 1.4$, $s_q = 1.2$, $s_\pi = 1.4$, and $s_m = 1.98$. The LBM simulation in this study was optimized using GPU computing and verified by comparing simulated values with laboratory-measured results. The model used in this article has been verified by Chen *et al.*[53]

In the continuum-surface-force (CSF) model, a body force is added to the evolution equations to create the local stress jump across the interface, which can be expressed as

$$\mathbf{F} = \frac{1}{2}\sigma\kappa\mathbf{C}, \tag{12}$$

where σ is the surface tension, and κ is the interface curvature, which can be calculated by $\kappa = [(\mathbf{I} - \mathbf{nn}) \cdot \nabla] \cdot \mathbf{n}$, where $\mathbf{n} = \frac{\nabla\phi}{|\nabla\phi|}$ is the normal direction vector of the fluid interface. $\mathbf{C}$ is the color gradient,

$$\mathbf{C} = \nabla\phi = \frac{3}{\delta t}\sum_i \omega_i \mathbf{e}_i \phi(t, \mathbf{x} + \mathbf{e}_i\delta t), \tag{13}$$

where ω_i is the weight coefficient for the D3Q19 lattice and the order parameter ϕ is defined by $\phi = \frac{\rho_r - \rho_b}{\rho_r + \rho_b}$.

Wetting model in this study combining the LB CSF-based color-gradient model and the geometrical wetting model. To ensure minimum spurious velocity near the three-phase contact line, there are two key steps. The first step is the extrapolation of ϕ to solid boundary nodes,[54] so that the color gradient at fluid boundary nodes is minimized according to Eq. (13), thus guaranteeing that there are no strong interactions between the fluid nodes and solid nodes. The second step is to apply the zero-interfacial-force scheme[55] by extrapolating the normal direction vector $\mathbf{n}$ of the fluid interface to solid boundary nodes as the calculation of the interface curvature in Eq. (10) at the three-phase contact line requires the value of $\mathbf{n}$ at solid boundary nodes. In such case, the force $\mathbf{F}$ in Eq. (12) is minimized at the three-phase contact line, which significantly reduces the spurious currents. As a result, the geometrical wetting models can maintain good accuracy for contact angles.

In flow simulations, fluid parameters need to be set to match the physical system. The Ohnesorge number (*Oh* number), which is influenced by fluid properties and geometry but remains independent of injection velocity, serves as a more effective parameter for describing flows where capillary, viscous, and inertial forces interact. As illustrated in Eq. (14), *Oh* number can be related to the capillary number ($Ca = \frac{\mu V}{\sigma}$) and Reynolds number ($Re = \frac{\rho V d}{\mu}$), and the Darcy velocity V ($\mathrm{m\,s^{-1}}$) is calculated by averaging the pore-scale flow velocity u over the entire domain, and μ(Pa s) is the dynamic viscosity. This approach eliminates the velocity term, making the dimensionless number solely dependent on fluid properties and geometry. Consequently, the Ohnesorge number offers a more accurate description of the flow, allowing for the viscosity ratio parameter to be used to control the type of fluid in the simulation

$$oh = \sqrt{\frac{\mathrm{Ca}}{Re}} = \frac{\mu}{\sqrt{\rho\sigma d}}, \tag{14}$$

where the velocity term disappears and the dimensionless numbers depend solely on fluid properties and geometry. Here, σ ($\mathrm{N\,m^{-1}}$) is the surface tension, μ and ρ ($\mathrm{kg\,m^{-3}}$) are the dynamic viscosity and density of the displacement phase (gas phase in this study), respectively, and d (m) is the characteristic length of the pore geometry, which in this study refers to the mean pore diameter.

Model validation

We verify the simulated permeability by using the previous single-phase flow experimental data[56] (experiment 1). The microfluidic experimental setup is depicted in Fig. 3. In this experiment, once the microfluidic model was fully saturated with water, air-free water was continuously injected at a flow rate of 100 μl/min. The stable pressure value was recorded, and the permeability of the microfluidic model was calculated accordingly. The experiment involved four groups of microfluidic models with porous structures arranged in a square arrangement, with micropillar diameters of 500, 1000, 1500, and 2000 μm. Fluid simulations were also performed on these structures, and the simulated values closely aligned with the experimental results [Fig. 4(a)], demonstrating the high accuracy of the single-phase flow simulations.

Then, we conducted two-phase microfluidic experiments involving air displacing water (experiment 2). In this experiment, the

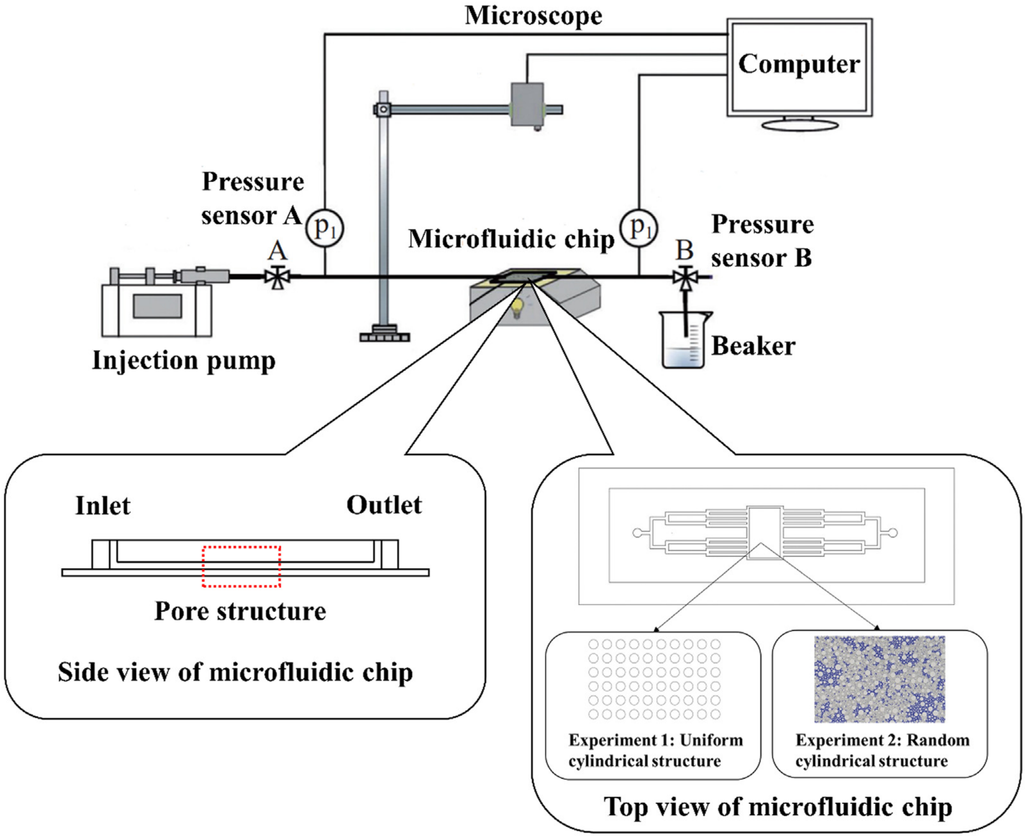

FIG. 3. Schematic diagram of the microfluidic chip experimental setup.

microfluidic model was initially completely saturated with dyed water. Air was injected into the model at a rate of 100 μl/h. When the two-phase distribution in the microfluidic model reaches a quasi-steady state, images are captured using a microscope to calculate the gas saturation. Six groups of microfluidic models were set up, each featuring a porous structure with a randomly arranged cylindrical pattern. As shown in Fig. 4(b), the comparison between the experimental and simulated gas saturations had an error of less than 5%, demonstrating the reliability of the two-phase flow numerical model. Additionally, by comparing the numerical simulations with the experimental results, we observed that the simulations successfully captured the main flow behaviors observed in the experiments. The above results illustrate the effectiveness of the two-phase flow model, which can be applied to subsequent methane emission studies from soil.

Parameter settings for flow simulation

Through these experiments, we effectively verified the reliability of both single-phase and two-phase flow simulations. Following this, we conducted two-phase flow (methane and water) simulations on the undisturbed soil samples. The methane properties used in this study include a dynamic viscosity of 0.011 mPa s, a density of 0.701 kg m^{-3},[57] and a surface tension of 71.64 mN m^{-1} at the methane–water interface.[58] The viscosity of water is 0.893 mPa s.[59] Due to methane's insolubility in water, the simulation does not account for methane dissolution.

The average methane emission rate measured in rice field cores throughout the growing season was 16 mg m^{-2} h^{-1}.[60] Based on this, we applied a constant velocity boundary condition of 16 mg m^{-2} h^{-1} at the inlet, a convective boundary condition at the outlet, and a no-slip boundary condition on the four lateral sides as well as the interior solid surfaces. The dimensionless parameters used in the simulation were converted from the above-mentioned *Oh* numbers. The 3D computational area is in a resolution of 40 μm per grid. One pixel corresponds to one LB lattice unit. Additionally, the conversion between time and velocity is carried out using the capillary number (*Ca*).

This model simulated the process of methane emission from the soil under flooded conditions, with the gas inlet positioned at the bottom of the soil in the vertical direction. Since methane is primarily produced in the rhizosphere,[61] we selected the calculation domain of $300 \times 300 \times 300$ around the roots. By simulating the methane saturation process, we aimed to elucidate methane emissions from soil pores. The level of saturation is quantified by the dimensionless number V_p, which represents the ratio of the volume of injected gas (cumulative volume) to the overall pore volume. For example, a V_p value of 0.25 means that the injection of methane gas is 25% of the total pore volume, while a V_p of 1 signifies that the methane gas volume injected is equal to the total pore volume. Methane saturation (*Sa*) is defined as

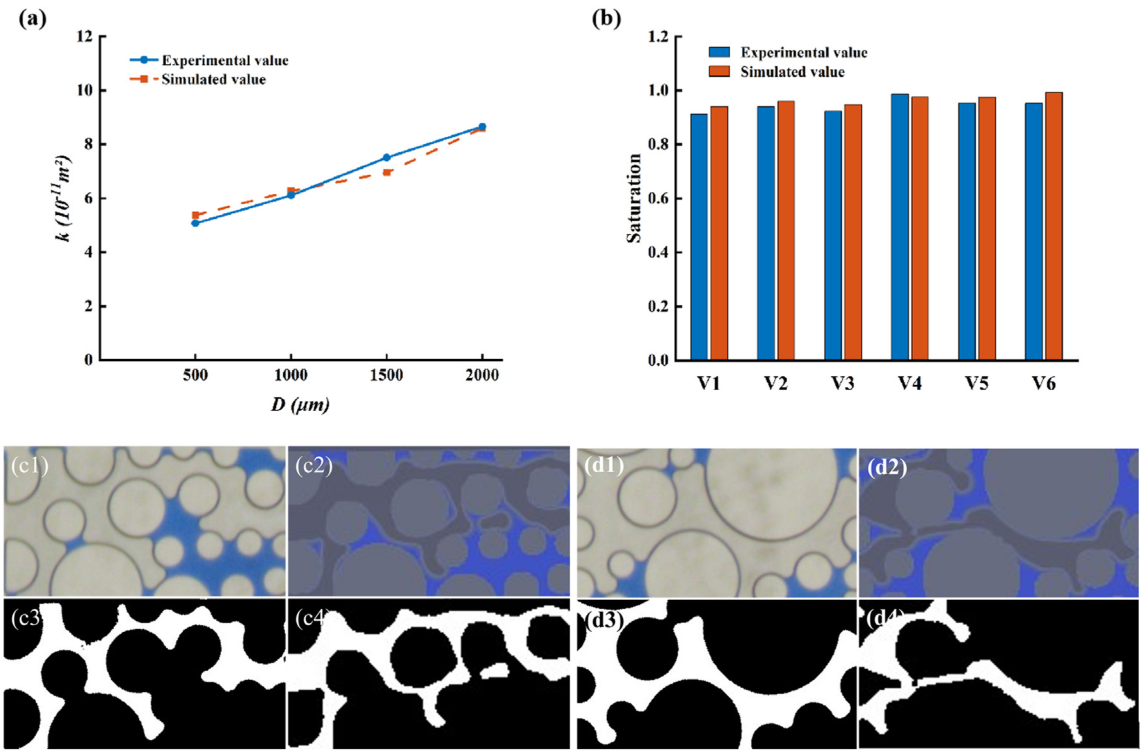

FIG. 4. (a) Experiment 1: comparison of experimental and simulated permeability values for the chip model with micropillars arranged in a square pattern at a porosity of 0.60; (b) experiment 2: comparison of experimental and simulated gas saturation values across six microstructures. Gas distribution images in the microfluidic chips for (c) sample 1 and (d) sample 2: (c1) and (d1) experimental results; (c2) and (d2) simulation results; (c3) and (d3) binary images of the experimental results; and (c4) and (d4) binary images of the simulation results (where the white areas represent the distribution of air).

the ratio of the gas volume in the sample pores to the overall pore volume. On the other hand, to illustrate the impact of porosity, Sa_T is defined as the ratio of the gas volume in the sample pores to the total volume of the sample.

RESULTS AND DISCUSSION

Pore and hydraulic characteristics

By analyzing the CT images of each soil sample, we measured the pore parameters using the morphological method, quantifying the pore size distribution and connectivity of the soil. Figure 5 presents the pore size distribution for both rhizosphere and non-rhizosphere soils.

The pore size distributions of the two non-rhizosphere soil samples (RL1 and RL2) are quite similar, with the most frequent pore size centered at 140 μm. In contrast, the rhizosphere soil samples (R1 and R2) show a broader distribution, with an increase in the proportion of larger pores and a decrease in the proportion of smaller pores, particularly around 140 μm. This notable difference in pore size distribution between rhizosphere and non-rhizosphere soils is linked to the influence of rice roots.

In addition to analyzing the pore size, we also calculated parameters such as porosity and critical pore diameter to provide a more

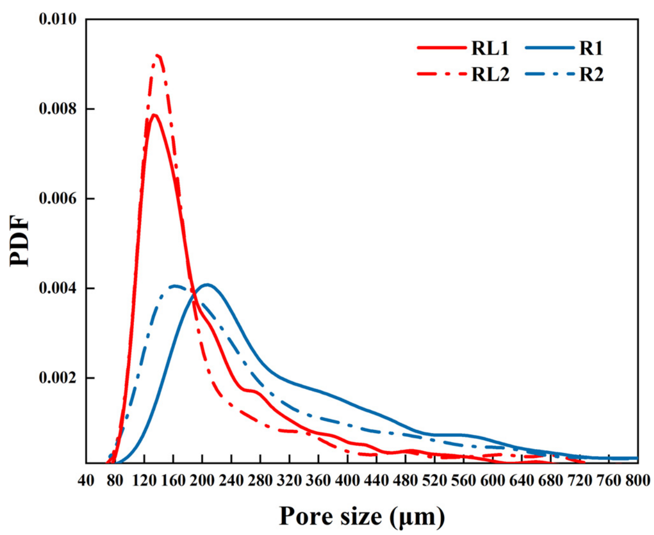

FIG. 5. Pore size distribution of each soil sample (RL1 and RL2 represent non-rhizosphere soil samples, while R1 and R2 represent rhizosphere soil samples).

TABLE I. Pore characteristics of soil cores calculated from x-ray tomography data. φ represents porosity, E represents the connectivity using the Euler number method, and Γ represents the tortuosity.

Soil name		φ (%)	Critical diameter (μm)	E	Γ
Non-rhizosphere soil	RL1	25.9	226	98	2.14
	RL2	23.8	235	57	2.41
Rhizosphere soil	R1	12.3	335	−162	1.58
	R2	10.1	255	−40	1.16

comprehensive evaluation of the pore structure in the soil samples (Table I).

Previous studies have indicated that the interaction between soil and plant root systems is influenced by the combined effects of both soil and plant conditions.[27] Our findings revealed that non-rhizosphere soils exhibit higher porosity levels compared to rhizosphere soils, suggesting a reduction in porosity in the rhizosphere due to the presence of rice roots. This aligns with the work of Daly *et al.*,[62] who showed that rhizosphere soil porosity is considerably lower than bulk soil, based on porosity measurements derived from CT scan data. Additionally, research has demonstrated that plants with fibrous root systems typically experience a 22%–24% reduction in rhizosphere porosity.[14] Soil pore structure exerts a significant influence on the interactions between roots and soil. When roots can grow into a highly connected macropore system with high connectivity, the rhizosphere is more porous compared to the bulk soil. Conversely, in soils with poor connectivity, roots tend to compact the soil. Experimental evidence has shown that within a region hundreds of micrometers from the root, compaction of the rhizosphere was observed for intermediate bulk density along with relatively high root length densities, resulting in a decrease in visible porosity by more than 25%.[15] In the low connectivity non-rhizosphere soil of this study, roots lack sufficient pore space for growth and thus compact the soil, leading to a marked reduction in porosity.

The critical pore diameter d_c, defined as the diameter of the largest sphere that can pass through the pore network, is a significant parameter influencing the hydraulic conductivity of soil. In our study, the critical diameters of two rhizosphere soil samples, R1 and R2, were found to be 335 and 255 μm, respectively. These values are larger than the critical diameters of non-rhizosphere soil samples, which range between 220 and 240 μm.

Differences in the pore network connectivity were observed between rhizosphere and non-rhizosphere soils. The Euler numbers for the two non-rhizosphere soil samples were positive (98 and 57), while those for the rhizosphere soil samples were negative (−162 and −40), indicating better pore connectivity in the presence of roots. This suggests that plant roots enhance soil connectivity by altering the pore structure. This observation aligns with the findings of Bacq-Labreuil *et al.*,[49] who proposed that roots influence the size distribution and connectivity of soil pores by creating, using, and occupying pore spaces. Lucas *et al.*[15] observed that as root length density increases, the Euler number decreases, which suggests an enhancement in soil connectivity within the rhizosphere. Additionally, the analysis of tortuosity across rhizosphere and non-rhizosphere soils revealed that non-rhizosphere soil exhibits higher tortuosity.

The above results show the differences in pore structure and hydraulic properties between rhizosphere soil and non-rhizosphere soil. Given that all samples were collected from the same region, these differences are primarily attributed to the presence of rice roots. Whalley *et al.*[63] suggested that plant roots can interact with soil through physical, chemical, and biological mechanisms, such as compaction, secretion release, and enhanced soil aggregation. Of these, the physical processes of the root–soil interaction, including macropore formation, soil particle rearrangement, and pore blockage, are likely responsible for the notable variations in porosity, pore distribution, connectivity, tortuosity, and permeability observed in our study.[64] To further elucidate the impact of rice roots on soil, we also analyzed the pore structure of the soil adjacent to the roots and observed significant pore occupation and compaction. Figure 6(a) illustrates the condition of pore occupation by roots. The occupation of pores by roots results in a reduction in pore size and a decrease in porosity. Figure 6(b)

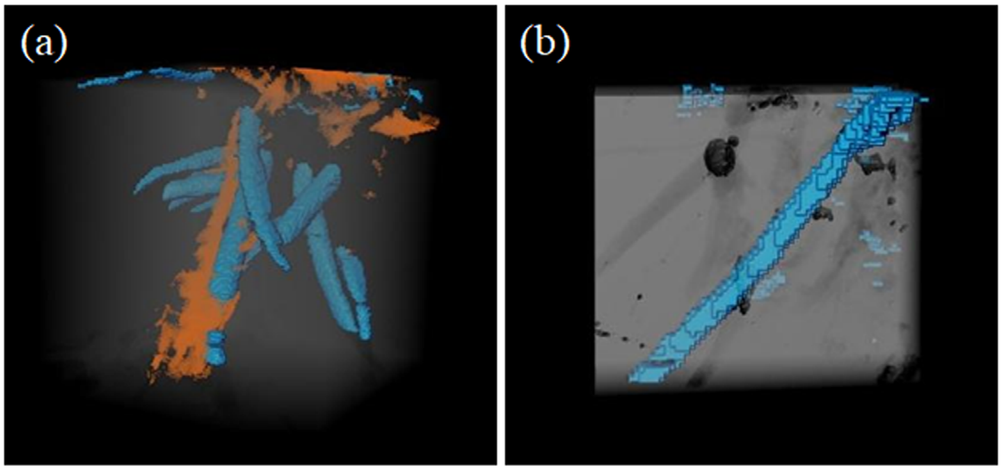

FIG. 6. The effects of roots on soil (a) occupation and (b) compaction. (Blue represents roots, and yellow represents pores.)

depicts the compaction effect, where roots compress the surrounding pores, leading to the collapse of the original pore structure. Additionally, the re-orientation of soil particles induced by roots results in the formation of larger pores and an increase in the critical diameter. Root shrinkage or decay may also contribute to the development of larger pores. Given that roots tend to grow into pores, they can create pathways between previously non-connected pores. This enhances the connectivity of rhizosphere soil and potentially makes fluid pathways more direct, which corresponds to the observed reduction in the tortuosity of rhizosphere soil.

Effect of pore characteristics on methane emissions

Under flooding conditions, methane gas displaces water in the soil. The initial and displaced states are illustrated in Figs. 7(a) and 7(b), respectively.

We generated methane saturation (*Sa*) curves for RL1, RL2, R1, and R2 as a function of the volume of injected methane [see Fig. 8(a)]. The saturation curves for the RL1 and RL2 samples exhibited similar behavior. As the volume of injected gas increased, the *Sa* of both RL1 and RL2 soil samples also increased. However, when the injected gas volume reached approximately 0.35, *Sa* no longer rose with additional gas injection. In contrast, at the same injected gas volume, the Sa values for the R1 and R2 soil samples were lower. The final *Sa* of the R1 samples remained at 0.21, while the R2 samples stabilized at 0.26, suggesting that methane is more likely to escape from rhizosphere soil under the same injected gas volume.

Due to the influence of plant roots (R1 and R2), the methane saturation in rhizosphere soil samples decreased, suggesting that methane diffuses more easily through rhizosphere soil. In addition, R1 exhibits a higher proportion of large pores compared to R2 (Fig. 5), leading to the lowest saturation. To demonstrate the effect of soil pore size more clearly on methane saturation, we categorized soil pores according to pore size. Kim *et al.*[65] divided soil pores into coarse mesopores (200–1000 μm) and macropores (>1000 μm). Based on the pore distribution in paddy soil and the resolution limits of CT, we classified the pores into small pores (80–200 μm) and large pores (>200 μm), as shown in Fig. 8(b). The non-rhizosphere soil samples were characterized by a predominance of small pores, with a relatively uniform proportion of these small pores. In contrast, the rhizosphere soil exhibited a lower proportion of small pores, which was associated with a corresponding decrease in methane saturation. Additionally, the critical diameter of rhizosphere soil is larger than that of non-rhizosphere soil, while the former exhibits a lower methane saturation. A clear negative correlation is observed between critical diameter and methane saturation. In this study, the critical diameter of the four samples also shows a strong association with the proportion of small pores; the smaller the proportion of small pores, the larger the critical diameter of the soil samples. Both the proportion of small pores and the critical diameter essentially reflect the pore size distribution. The relationship between soil pore size and methane retention may necessitate further elucidation.

Apart from the previously mentioned pore size, connectivity and tortuosity are also key parameters for soil pore. It is essential to systematically investigate how these factors influence methane saturation. Based on the reconstructed soil structure images (Fig. 2), it is evident that non-rhizosphere soil, despite having a positive Euler number, still contains some connected pores. These connected pore structures facilitate methane ingress but may also create pathways that are not connected to the upper layers of the soil core, which could play a positive role in methane retention. In contrast, rhizosphere soil samples exhibit markedly enhanced connectivity, which promotes methane emission. Additionally, the growth of roots in the soil may increase the number of connected pathways in rhizosphere soil, leading to a decrease in tortuosity and facilitating methane emission. Conversely, microchannels within non-rhizosphere pores are more winding, requiring methane gas to travel longer paths in these areas, thereby exerting a stronger retention effect on methane. Soil pore connectivity determines the proportion of pores through which fluid can flow, while tortuosity acts as a partial impediment to fluid movement.

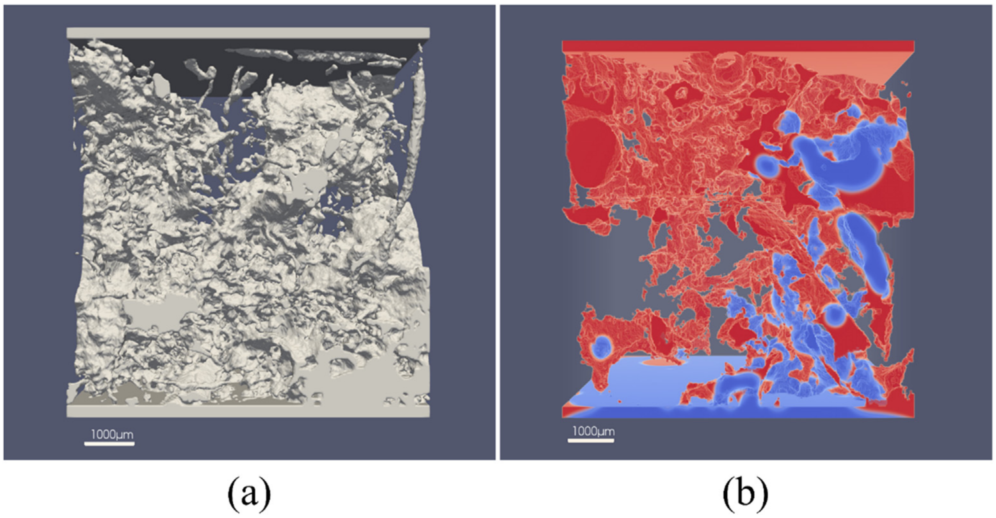

FIG. 7. Two-phase flow (methane and water) simulation processing: (a) setting buffer zones at the entrance and exit of the pore structure, and (b) simulating methane emissions, where red represents water and blue represents methane.

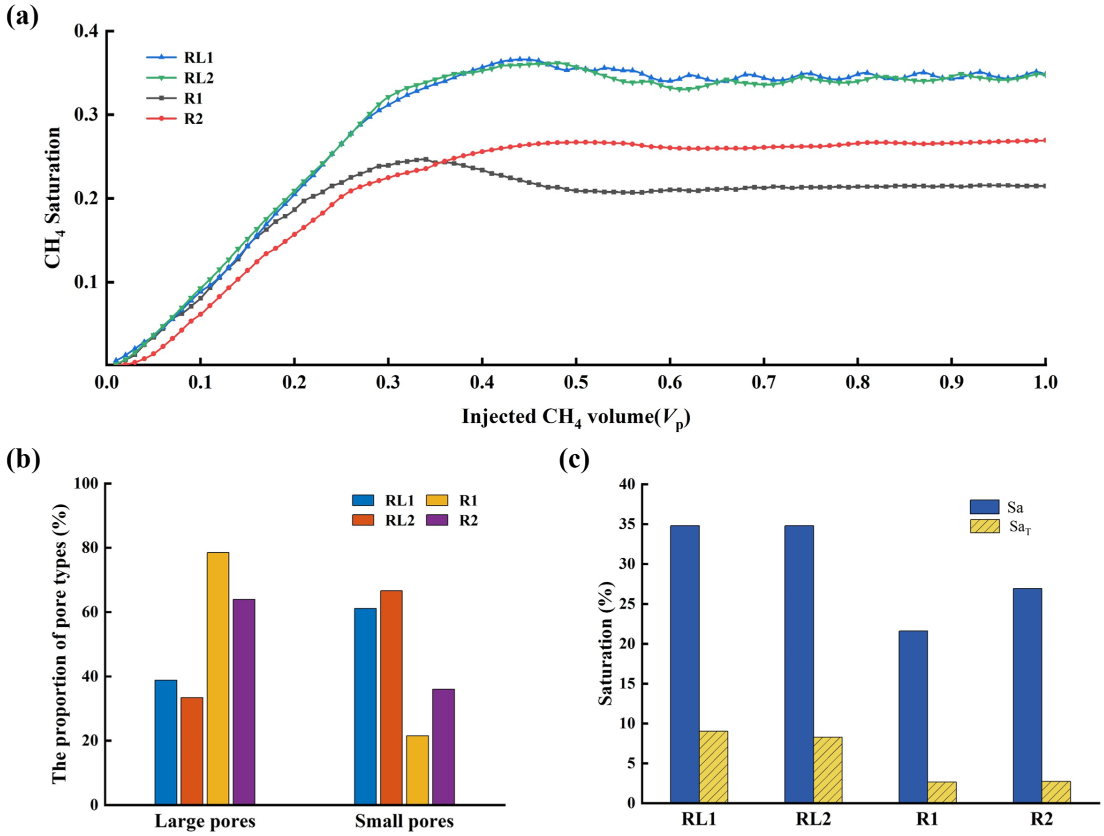

FIG. 8. (a) Relationship between the volume of methane entering the soil (injected CH_4 volume) and methane remaining in the soil (CH_4 saturation), (b) the proportion of the two types of pores in each soil sample, and (c) the methane saturation of the soil sample, considering and not considering the contribution of porosity (where *Sa* is the ratio of the gas volume in the pores to the overall pore volume, and Sa_T is the ratio of the gas volume in the pores to the total volume of the sample).

These results indicate that the combination of poor connectivity and high tortuosity contributes to the increased methane retention capacity of non-rhizosphere soils compared to rhizosphere soils. In soil pore structures with a high proportion of small pores and low critical diameter, more methane is retained, leading to a higher saturation level.

In this study, the calculated methane saturation is relative to the overall pore volume, thus not accounting for the influence of porosity. For soil samples of equal volume, we introduced the Sa_T parameter to incorporate the effect of porosity. As shown in Fig. 8(c), the low porosity of rhizosphere soil leads to a further increase in the difference in saturation between rhizosphere and non-rhizosphere soils when the porosity parameter is considered. Consequently, the methane retention in rhizosphere soil is lower than that in non-rhizosphere soil. These findings indicate that modifying the soil pore structure can significantly affect methane emissions.

To better understand the methane retention effect in different regions of the soil, we simulated the methane emission processes in the collected soil samples, as shown in Fig. 9. As the amount of methane gas increases, the soil pores gradually become saturated. When the ratio of the generated gas volume to the total pore volume reaches 0.22, primary gas emission channels have already formed in the rhizosphere soil sample R1, while narrower pathways appear in the rhizosphere sample R2. In contrast, the gas flow paths in RL1 and RL2, which exhibit high tortuosity, are more complex. Multiple methane emission pathways form in the non-rhizosphere soil, but no methane has been released at this stage. When the methane gas volume increased to 0.27, the main emission channels also form in the rhizosphere soil sample R2. As the methane gas volume continues to rise, eventually filling the pores, emission channels are observed in all soil samples.

Notably, non-rhizosphere soils develop several potential methane emission pathways. However, due to poor connectivity, some pathways within the soil may not be connected to the top of the soil core. These disconnected pathways are unable to release methane, resulting in methane retention. In contrast, the rhizosphere soil, although it has fewer emission channels, benefits from high pore connectivity, which facilitates more efficient methane escape. It is important to note that at $V_p = 0.22$, despite both R1 and R2 being rhizosphere soil samples, only R1 can form emission channels, while R2 cannot (Fig. 9). This difference arises from variations in their parameters: R1 has higher connectivity, the lowest proportion of small pores, and the highest critical pore diameter, making methane escape easier, despite its higher tortuosity. Therefore, it can be inferred that in our study, tortuosity has a lesser impact on methane emission compared to other factors. The observation that R1 forms emission channels more readily than R2

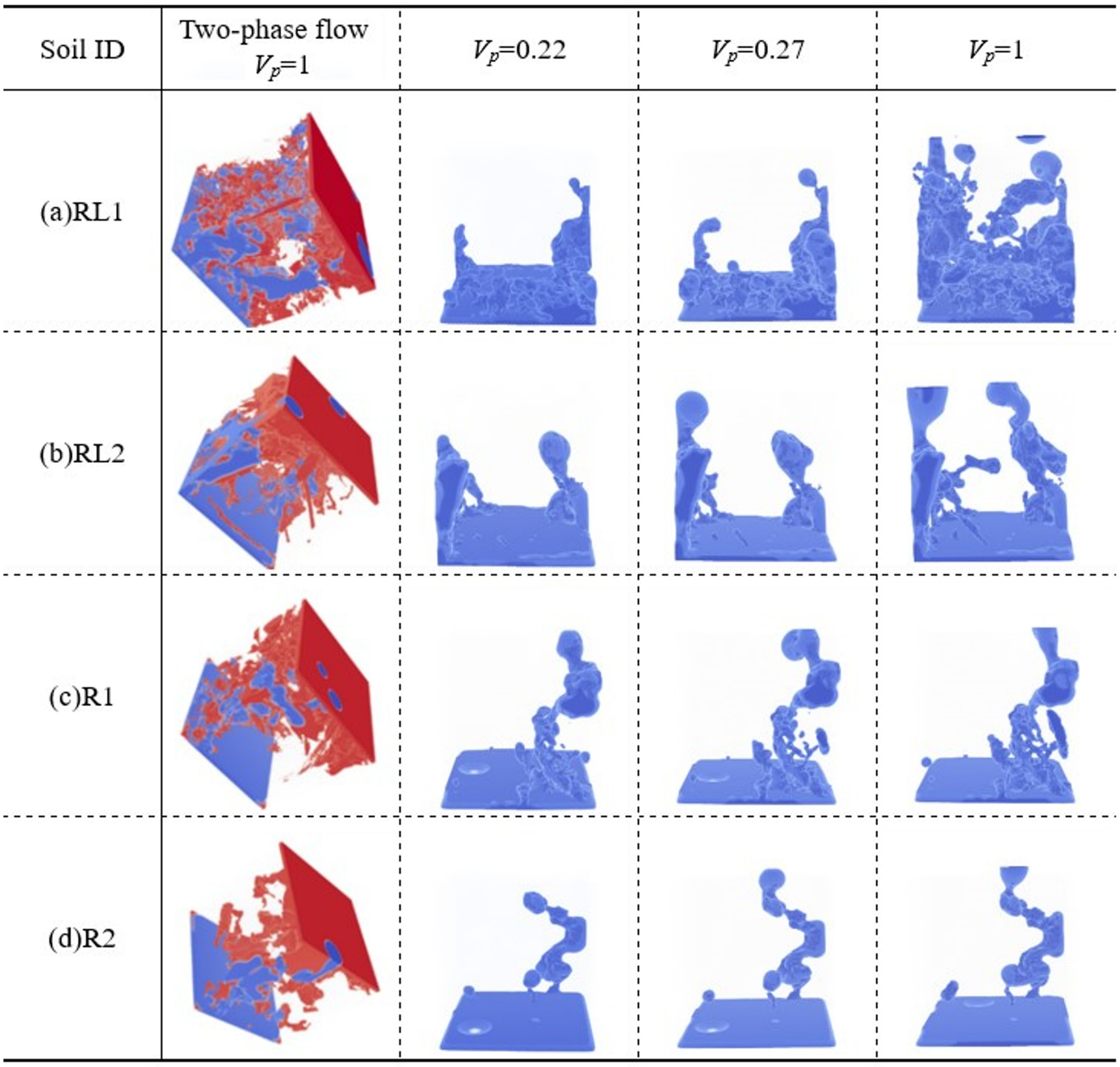

FIG. 9. Methane emission simulations for each soil sample. RL1 and RL2 represent the non-rhizosphere soil samples, while R1 and R2 correspond to the rhizosphere soil samples. $V_p = 1$ indicates that the methane gas produced in the rhizosphere is equal to the total pore volume, whereas $V_p = 0.22$ signifies that the methane gas produced accounts for 22% of the total pore volume. In the diagram, the blue phase represents emitted methane gas, while the red phase denotes water within the pores.

aligns with the methane saturation curves, where R1 exhibits lower methane saturation (*Sa*) than R2. It is readily understood that in highly connected soil pore structures, methane can easily escape, which is detrimental to methane retention. The mechanism by which pore size affects methane retention is not yet clearly understood. Consequently, we attempted to link the flow paths of methane in two-phase flow simulations with the spatial distribution of pore sizes. Here, we mapped the spatial distribution of large and small pores in the soil to describe pore size distribution (Fig. 10). It can be observed that small pores (in purple) are distributed without a specific pattern in the soil and tend to cluster in groups between large pore regions. For RL1 and RL2 samples, many small pores are present in the main channels, which facilitates methane entry into the surrounding areas, resulting in higher methane saturation (white circle). In contrast, R1 and R2 samples have fewer small pore regions, causing methane to directly escape from the main channels, resulting in lower methane saturation. This can also be explained from the perspective of critical diameter: non-rhizosphere soil samples have smaller critical diameters, and the main channels pose a significant barrier to methane emission, allowing methane to enter a large number of microchannels in the soil, whereas rhizosphere soil samples have larger critical diameters, enabling methane to easily escape from the main channels. Additionally, in RL1 and RL2 samples, we observed regions where methane was trapped, surrounded by a large number of small pores (green circle). These results indicate the positive impact of small pores on methane retention. Moreover, as shown in Fig. 10(a1), some pores (orange circle) are not connected to the top of soil core. These pores act as traps for methane [Fig. 10(a2)]. This finding intuitively illustrates the impact of connectivity on

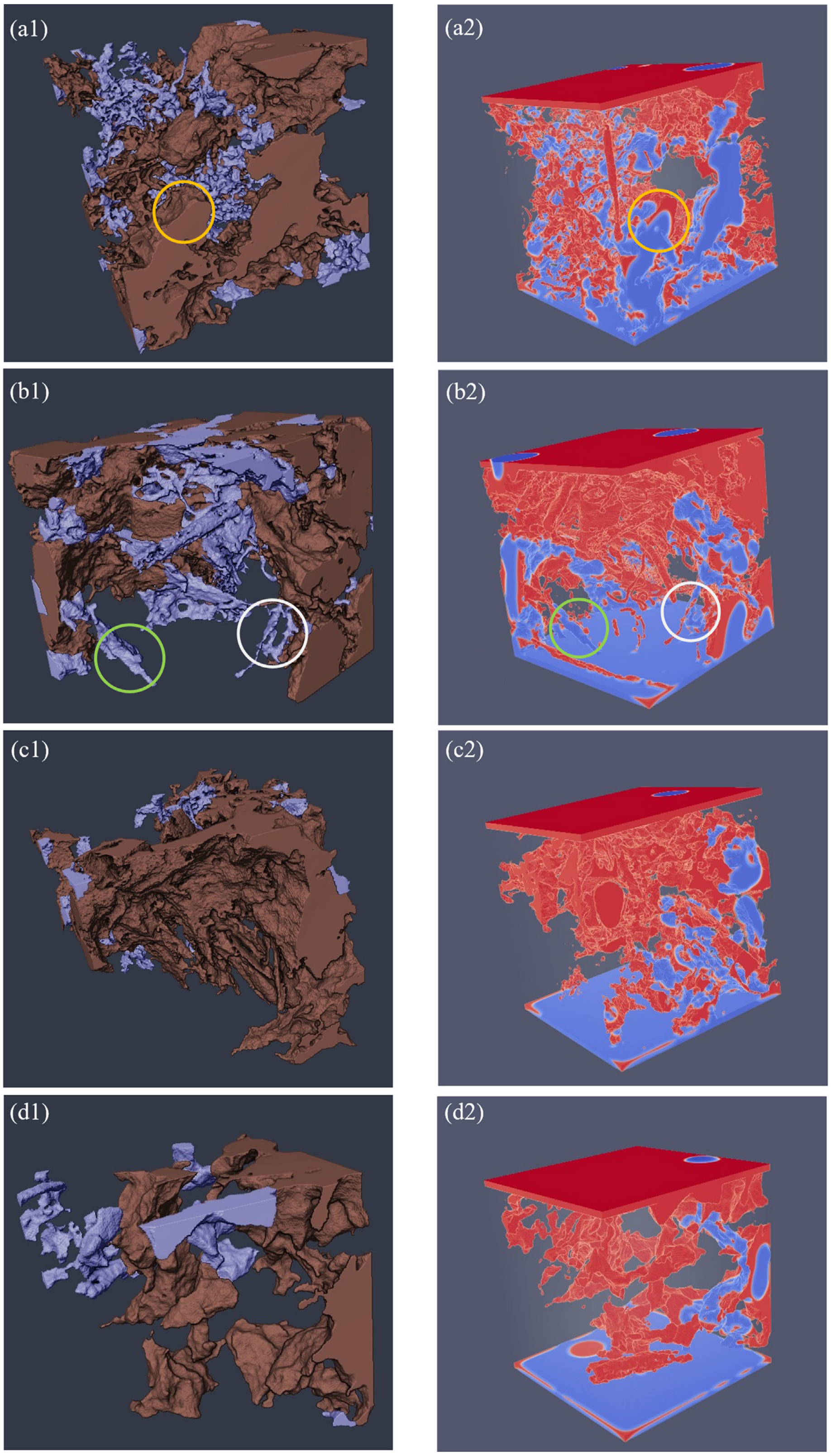

FIG. 10. 3D visualization of the spatial distribution of large (in brown) and small (in purple) pores (a1)–(d1), along with methane emission simulation (a2)–(d2) for (a) RL1, (b) RL2, (c) R1, and (d) R2. (Orange circle marks poorly connected dead-end pores; white circle marks small pores within the main channels; and green circle marks the region where methane is trapped in small pores.)

methane retention. Specifically, soils with poor connectivity tend to have more pathways that are not connected to the top of the soil core, thereby trapping methane.

Our study has certain limitations. While we modeled methane emissions within the pores of paddy soil, we did not consider the biochemical differences between rhizosphere and non-rhizosphere soils. Our primary aim was to assess the impact of the pore network structure on methane emissions from a physical perspective. This approach provides a foundation for future simulation studies that could incorporate more complex biochemical processes. Importantly, we observed clear differences in the pore structure between rhizosphere and non-rhizosphere soils, which subsequently affect methane emissions within these pores.

CONCLUSION AND IMPLICATIONS

By combining x-ray mCT and GPU-accelerated LBM simulation, our research revealed that rhizosphere soil exhibits distinct pore structure and hydraulic properties compared to non-rhizosphere soil, highlighting the significant impact of root systems on soil physical characteristics. As the root system expands, it can occupy and compact soil pores, resulting in a decrease in porosity, a reduced proportion of small pores, and an increase in the critical pore diameter. At the same time, this process significantly enhances connectivity and reduces tortuosity.

We performed two-phase flow simulations to obtain methane saturation results for both rhizosphere and non-rhizosphere soils. Our findings show that soil methane saturation is correlated with pore structure parameters, including connectivity, tortuosity, and, most importantly, pore size distribution, which plays a crucial role. An increase in the proportion of small pores and a decrease in critical pores and soil connectivity, along with an elevation in tortuosity, contributes to enhanced methane retention.

This study focuses on the pore-scale structure of rhizosphere and non-rhizosphere soil samples from flooded rice fields, systematically exploring the influence of the soil physical structure on methane retention. Additionally, the method demonstrated in our study can be applied to various pore structures, and through up-scaling, can further explore the impact of spatial and temporal variabilities in soil parameters on methane emissions. By integrating these findings with macro-level observations of methane emissions, we can refine our approach and develop effective strategies for mitigating soil methane emissions.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Wujun Zhang: Formal analysis (equal); Investigation (equal); Visualization (equal). **Jun Man:** Methodology (equal); Software (equal). **Xiuling Yu:** Formal analysis (equal). **Liuwu Fu:** Software (equal). **Xianhua Zheng:** Writing – review & editing (equal). **Rou Chen:** Methodology (equal). **Yajuan Zhuang:** Validation (equal). **Jiangjiang Zhang:** Validation (equal). **Xiaoyu Liang:** Project administration (equal). **Hongxiang Zhou:** Conceptualization (equal); Investigation (equal); Supervision (equal); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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